Remediation Agent (Agent 2)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Remediation Agent converts high-severity security incident findings into actionable, multi-platform containment playbooks, automated isolation scripts, and long-term security hardening measures while enforcing safety approval gates for destructive commands.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Second node in Pipeline B, executing directly after IncidentTriageAgent.
- Context Consumption: Consumes state.triage_report, state.extracted_iocs, and state.findings.
- State Propagation: Writes state.remediation_plan containing step-by-step mitigation commands.
- Human-in-the-Loop Safety: Enforces mandatory approval gates for commands that disrupt production services.

2. Multi-OS Command Generation Engine

- Linux CLI: Generates iptables firewall drop rules, ufw block commands, systemctl service termination, and pkill process isolation.
- Windows PowerShell: Generates Stop-Process, New-NetFirewallRule, and Disable-LocalUser scripts.
- Kubernetes Container Engine: Generates kubectl cordon node isolation, kubectl delete pod, and network policy restriction manifests.
- Cloud IAM & Infrastructure: Generates AWS CLI commands (aws ec2 revoke-security-group-ingress, aws iam detach-user-policy).

3. Destructive Action Flagging & Safety Gates

- Classification Rules: Commands that terminate active database connections, reboot servers, or delete storage volumes are flagged with is_destructive: true.
- UI Gate Enforcement: The frontend interface intercepts destructive actions and displays a modal requiring typing 'APPROVE' before copying or executing.
- Justification Audit Log: Every generated command includes a mandatory technical justification string for SOC compliance logging.

4. Output Schema & Playbook Structure

- RemediationAction Schema: Contains step_number, action_type (CONTAINMENT, ISOLATION, ERADICATION, RECOVERY), title, command, is_destructive, and justification.
- Long-Term Hardening: Provides architectural guidance for preventing recurrent exploitation vectors.